

OPTICAL FIBRE DEMO IN 'C'

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An optical communication system uses light as the carrier of information. Electromagnetic light waves propagate through the optical fibre in much the same way as radio signals propagate through the earth's atmosphere. Propagating light through the earth's atmosphere is difficult and often impractical. Consequently, optical fibres use glass or plastic fibre cables to contain the light waves, which are then guided properly and transmitted to the receiver end.

This program demonstrates various features of the fibre optic technology including physical structure of the optical fibre, light rays in the optical fibre, application of optical fibres in communication and various types of optical fibres.

Before proceeding to the software program, you may go through the following optical fibre fundamentals for a better understanding of the various figures included in the program.

Optical fibre

The term 'optical fibre' (or fibre optic) refers to the medium and the technology associated with the transmission of information as light pulses along a glass/plastic fibre about the diameter of a human hair. The fibres are arranged in bundles called 'optical cables' (refer Fig. 1) and used to transmit light signals over long distances.

An optical fibre is made of various layers including a transparent core surrounded by a transparent cladding material with a lower index of refraction (refer Fig. 2). The various layers are:

Core. Thin glass centre where the light travels

Cladding. Outer optical material surrounding the core that reflects the light back into the core

Buffer coating. Plastic coating that protects the fibre from damage and moisture

Jacket. The protective coating is surrounded by a buffer jacket which provides the cable additional protection against abrasion and shock



Fig. 1: Typical optical fibre cable

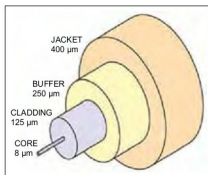


Fig. 2: Various layers in an optical fibre

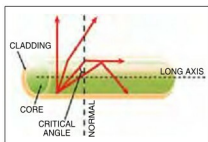


Fig. 3: Total internal reflection in an optical fibre



tion against abrasion and shock

Optical fibre link

A simplex optical fibre communication link consists of three elements: a transmitter, a receiver and the optical fibre cable.

Transmitter. The transmitter comprises a voltage-to-current converter, a light source and a source-to-fibre interface (light coupler). The light source is an infrared light-emitting diode (IR LED) or an injection laser diode (ILD). It can be modulated by a digital or analogue signal. The voltage-to-current converter serves as an electrical interface between the input circuitry and the light source and converts an input signal voltage into current that is used to drive the light source.

The source-to-fibre coupler (such as optical lens) is a mechanical interface. It couples the light emitted by the light source into the optical fibre cable.

Optical fibre cable. The fibre guide is the transmission medium, which is either an ultra-pure glass or a plastic cable. A regenerator can also be added for light amplification. It is basically spliced along the cable to boost the

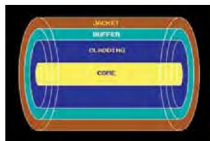


Fig. 4: Program output screenshot of optical fibre